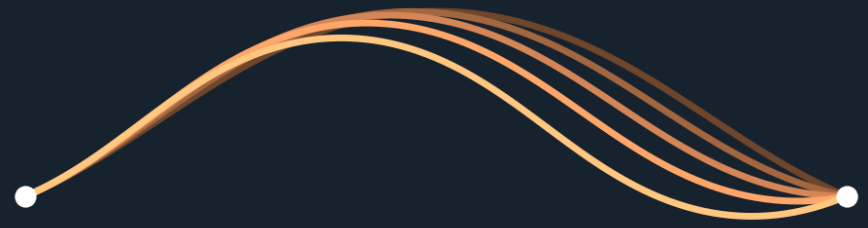
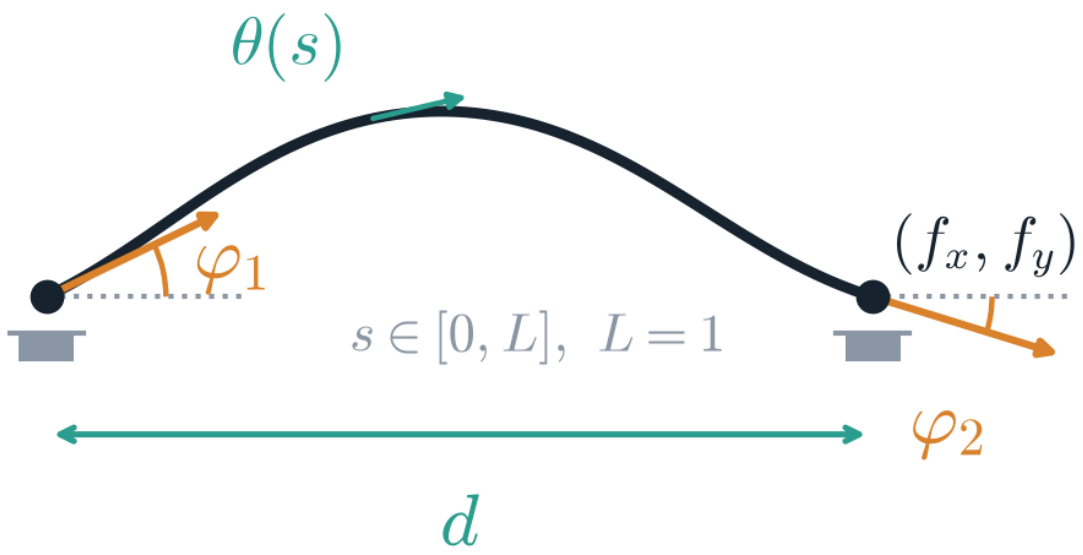
Mapping the clamped elastica across its snap surface

Continuation with buckling-mode branch tracking in FEniCSx / PETSc



One rod, three parameters, a map of equilibria



For every point (d, ϕ_1, ϕ_2) solve for the tangent angle θ(s) and the end forces f_x, f_y; record both clamp moments and the bending energy U.

d

End-to-end distance (L = 1)

0.70 – 0.99; Δd = 0.01

ϕ_1

Clamp angle at s = 0

$-\pi/2$ to $\pi/2$; $\Delta\phi = \pi/314 \approx 0.01$

ϕ_2

Clamp angle at s = L

Same range as ϕ_1; grid 30 × 315 × 315

Equilibria are stationary points of the bending energy

Bending energy

Inextensible Euler–Bernoulli rod, EI = 1

$$\Pi[\theta] = \frac{1}{2}\int_0^L \theta'(s)^2\, ds$$

Constraints

End position fixed, both ends clamped

$$\int_0^L \cos\theta\; ds = d, \quad \int_0^L \sin\theta\; ds = 0, \quad \theta(0) = \varphi_1, \quad \theta(L) = \varphi_2$$

Lagrangian

f_x, f_y enforce the two end-position integrals

$$\mathcal{L}[\theta, f_x, f_y] = \frac{1}{2}\int_0^L \theta'^2\, ds + f_x\left(\int_0^L \cos\theta\; ds - d\right) + f_y \int_0^L \sin\theta\; ds$$

f_x and f_y are the Lagrange multipliers of the end-position constraints — physically, the reaction forces at the right clamp. The study maps every stable stationary point of $\mathcal{L}$ the continuation reaches over the (d, ϕ_1, ϕ_2) box.

Stationarity gives the elastica equation and a mixed weak form

Euler–Lagrange equation

$$\theta'' + f_x \sin\theta - f_y \cos\theta = 0$$

$\theta(0) = \phi_1,\ \theta(L) = \phi_2$

1. $\theta \in$ P1 Lagrange on 200 elements, graded towards both clamps
2. $f_x, f_y \in$ "Real" space: one global DOF each
3. Clamp angles imposed as Dirichlet BCs on θ

Residuals assembled by FEniCSx (solver.py)

$$F_0 = \int_0^L \left(\theta' v' - f_x \sin\theta\ v + f_y \cos\theta\ v\right) ds = 0 \quad \forall v \in V_0$$

$$F_1 = \int_0^L (\cos\theta - d)\, q_x\, ds = 0$$

$$F_2 = \int_0^L \sin\theta\ q_y\, ds = 0$$

Post-processed per point

$$M_1 = \theta'(0), \quad M_2 = \theta'(L), \qquad U = \frac{1}{2}\int_0^L \theta'^{\,2}\, ds$$

Moments: end-element slope plus a half-element Taylor correction (second order)

Each parameter point is one Newton solve on 203 unknowns

$$J(\mathbf{u}^k)\,\delta\mathbf{u} = -\mathbf{F}(\mathbf{u}^k), \qquad \mathbf{u}^{k+1} = \mathbf{u}^k + \alpha_k\,\delta\mathbf{u}$$

u = (θ_0 … θ_{200}, f_x, f_y); ($0 < \alpha_k < 1$) from cubic backtracking line search

203

unknowns per point

201 nodal angles + 2 multipliers

LU

MUMPS direct solve

KSP preonly: exact linear steps

10^{-10}

relative tolerance

absolute 10^{-12}

50

max Newton its

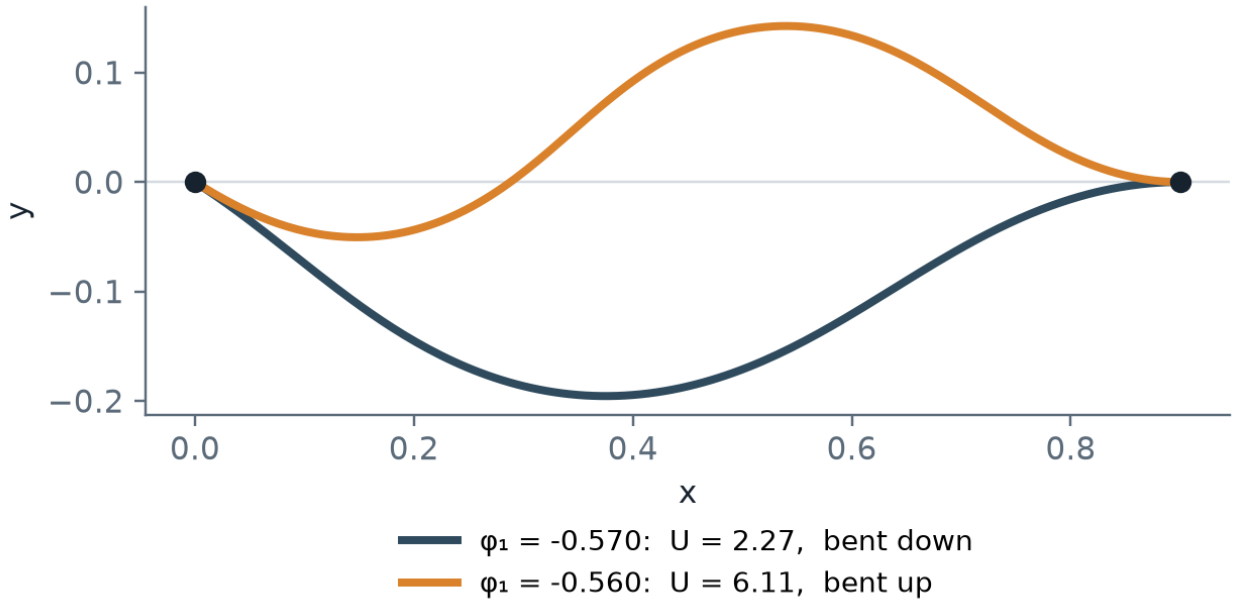
per warm start; more counts as a failure

At a fold of the solution branch, the Jacobian becomes singular, so the snap surface is exactly where the Newton solver fails, and where the continuation logic has to take over.

The snap surface

Where the equilibrium jumps from one branch to another

Two equilibria 0.01 rad apart: the rod has snapped



d = 0.90, φ₂ = 0, production run. Converged solutions on either side of the jump.

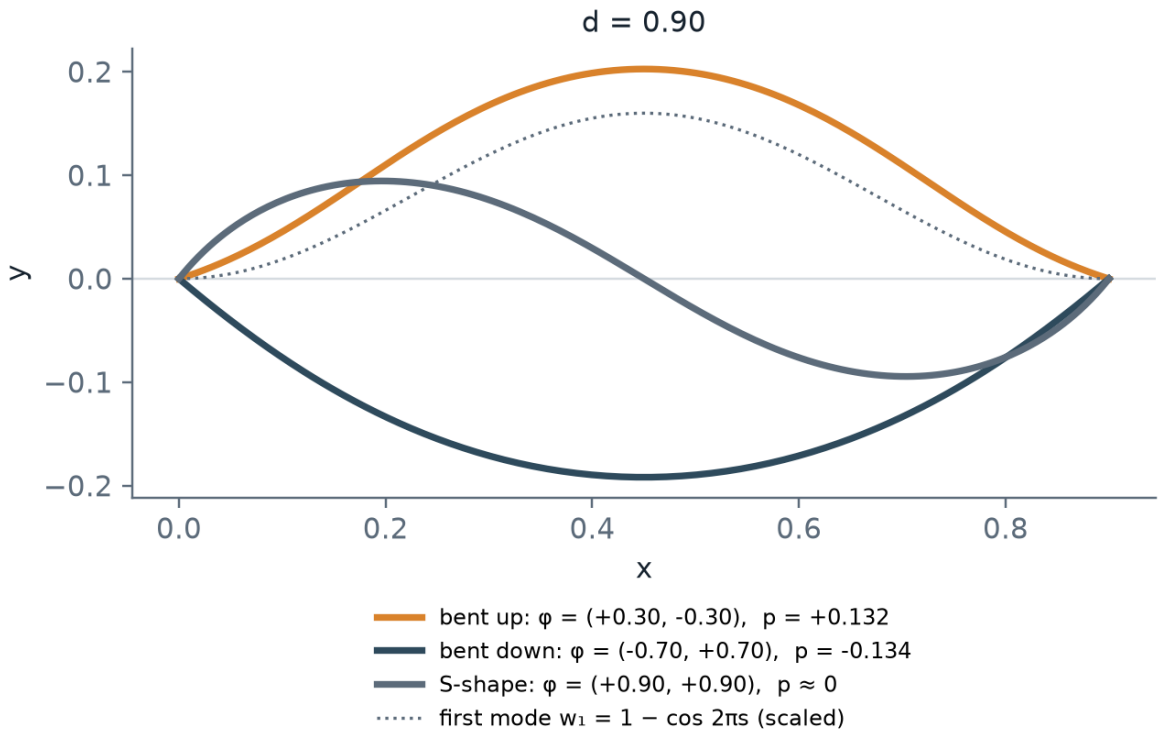
Why Newton struggles here

1. **Fold**
   The branch ends; the Jacobian goes singular and Newton steps blow up.

2. **Silent branch switch**
   A warm start from one side converges, but to the other branch.

3. **Residual can't tell**
   Both sides meet 10^{-10}, so convergence alone says nothing about which branch you are on.

Bent up or bent down: the first buckling mode decides



$$p = \frac{1}{\|w_1\|}\int_0^L y\ w_1\ ds$$

$$y(s) = \int_0^s \sin\theta\ dt, \qquad w_1(s) = 1 - \cos\frac{2\pi s}{L}$$

1 **First clamped buckling mode**

w_1 is the symmetric hump; the second mode is antisymmetric and carries no up/down information.

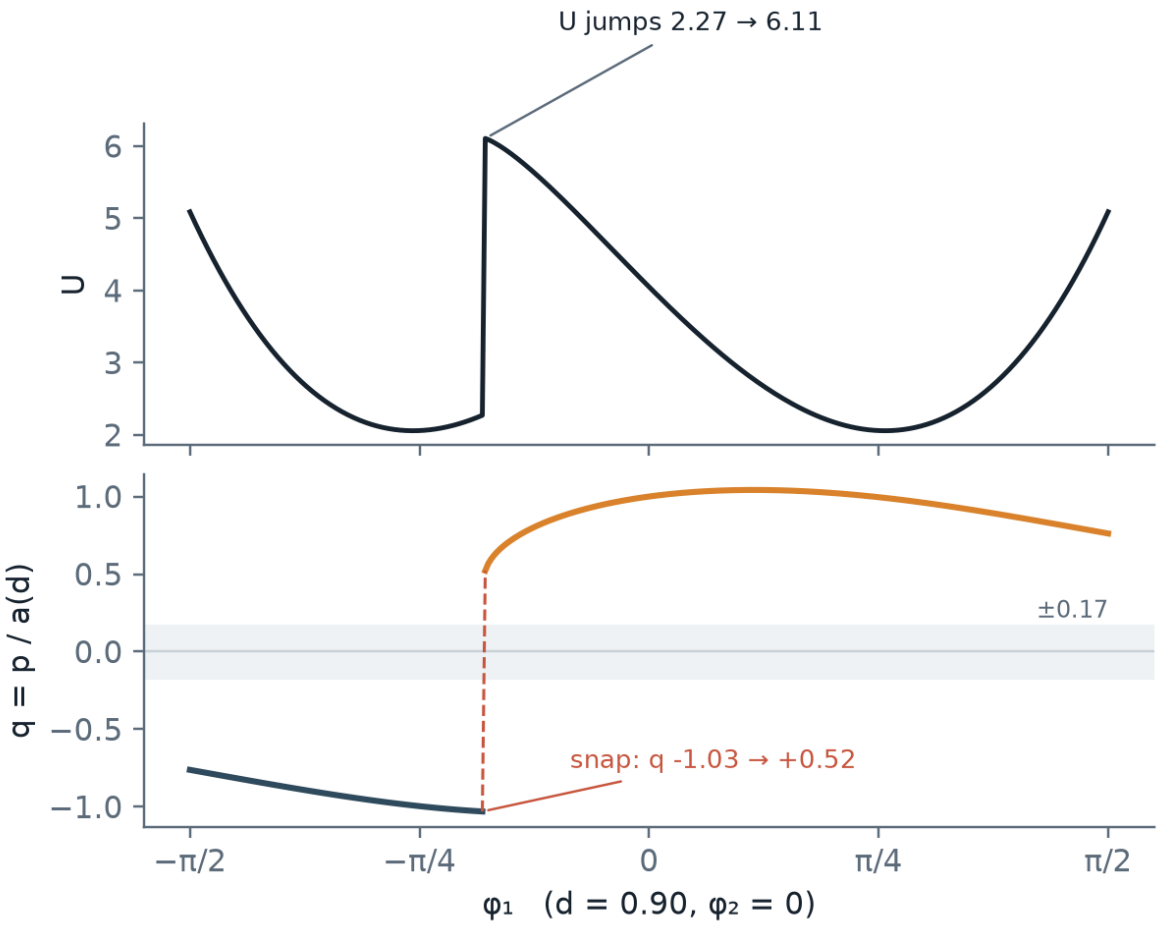
2 **Sign = branch**

$p > 0$: bent up. $p < 0$: bent down. The S-shape has $p \approx 0$.

3 **Built-in symmetry**

p is unchanged by the mirror $(\phi_1, \phi_2) \to (-\phi_2, -\phi_1)$ and flips under $(\phi_1, \phi_2) \to (-\phi_1, -\phi_2)$.

A snap is a sign change of p with a jump



$$q = \frac{p}{a(d)}$$

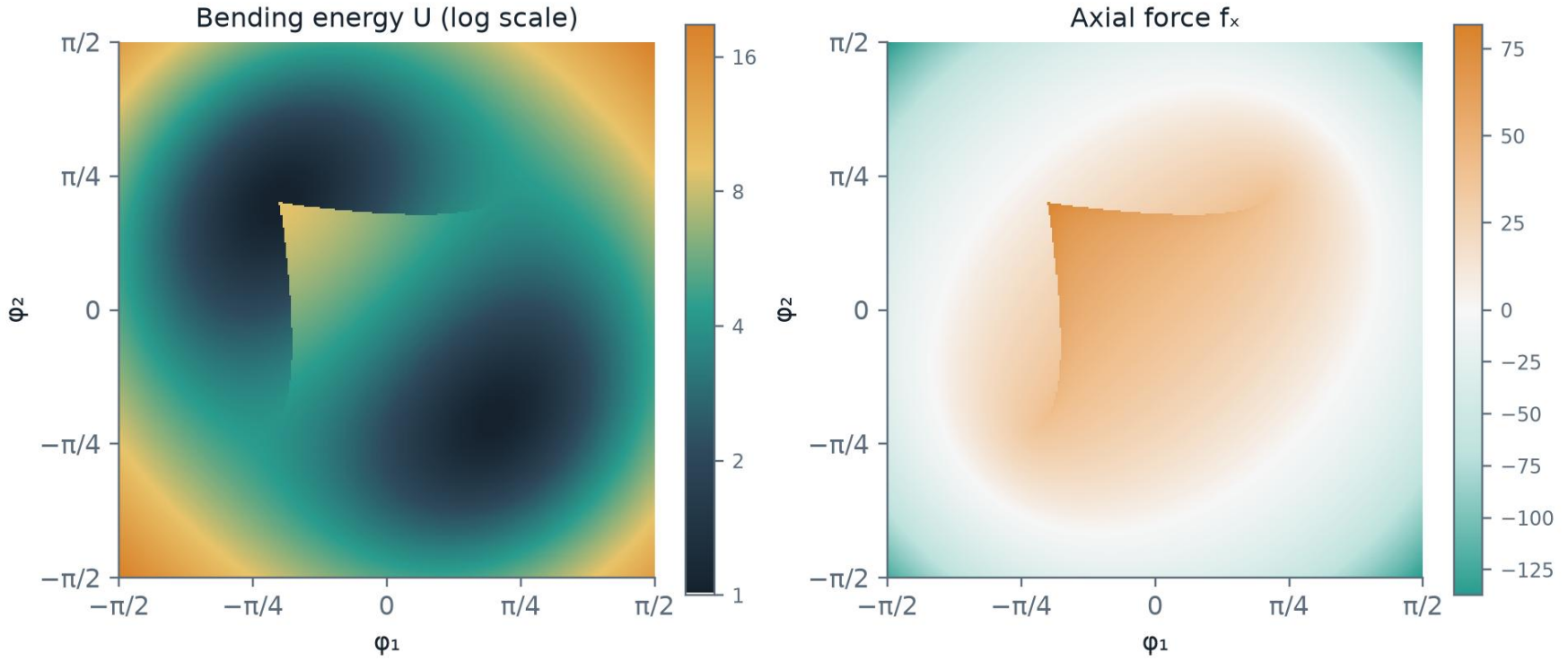
$$|\Delta q| > 0.17 \quad \text{and} \quad |\Delta q| > \frac{1}{2}\max(|q_a|, |q_b|)$$

1. **Scale out d**
   p shrinks like √(1 − d); a(d) is |p| of the exact first mode at (d, 0, 0), so q = ±1 there at every d.

2. **Snap**
   Sign change with a jump: q goes −1.03 → +0.52 in one 0.01 step. Smooth neighbours differ by about 0.003.

3. **Smooth crossing**
   Sign change without a jump, through the S-shape near $\phi_1 = \phi_2$. Not a snap.

Smooth fields, then a jump across an L-shaped surface



Production run, d = 0.90 layer: φ_1, $\varphi_2 \in [-\pi/2, \pi/2]$, $\Delta\varphi \approx 0.01$, 99,225 points.

Smooth interior

Grid neighbours differ by $|\Delta q| \approx 0.003$

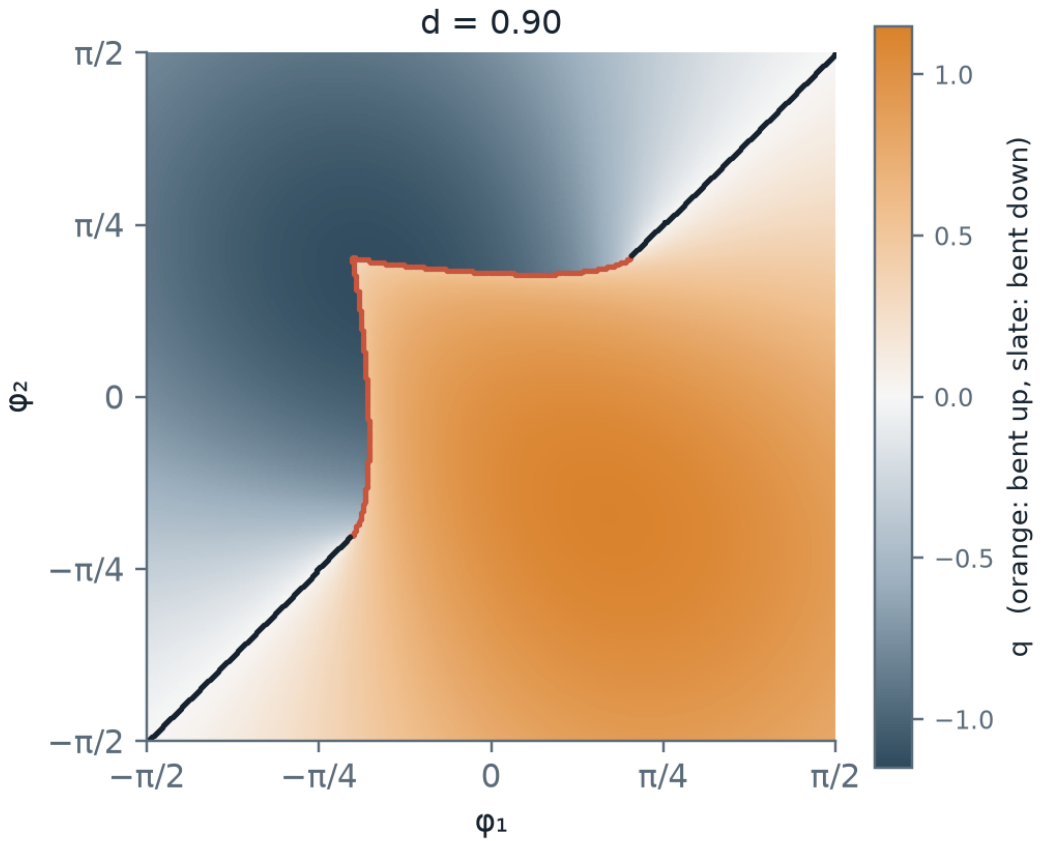
Sharp edge

Jump at $\phi_1 \approx -0.57$ and, by symmetry, $\phi_2 \approx +0.57$

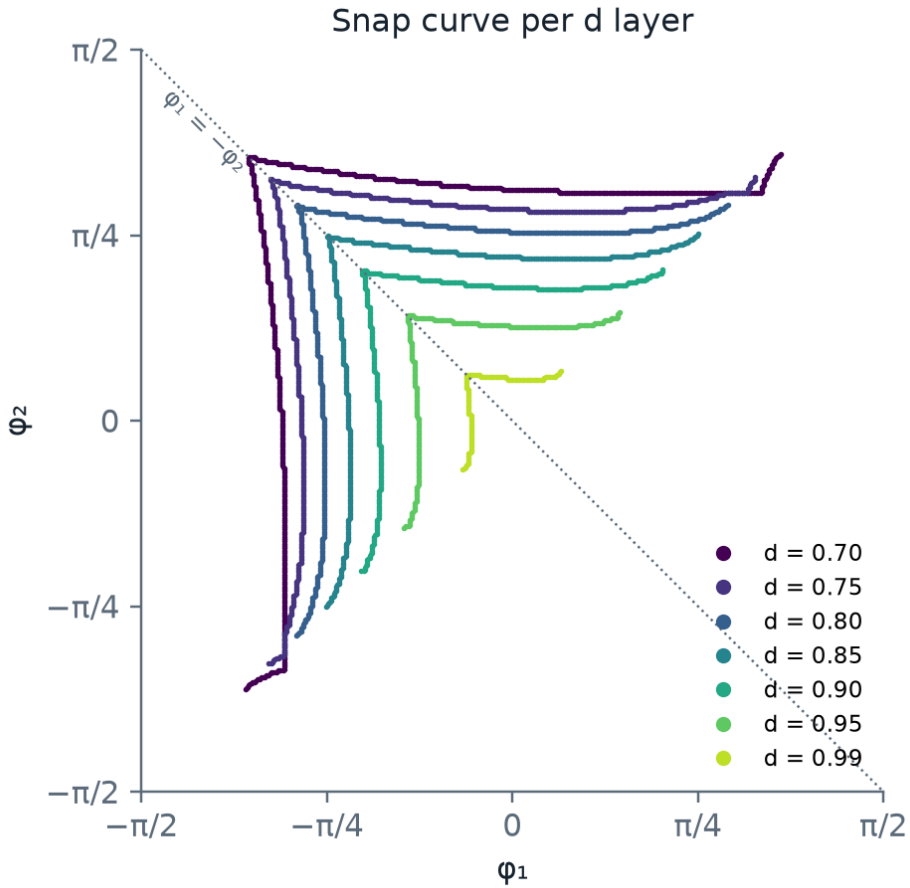
Large jump

U 2.27 → 6.11 and f_x 28 → 59 in a single 0.01 step

The sign of p traces the snap surface

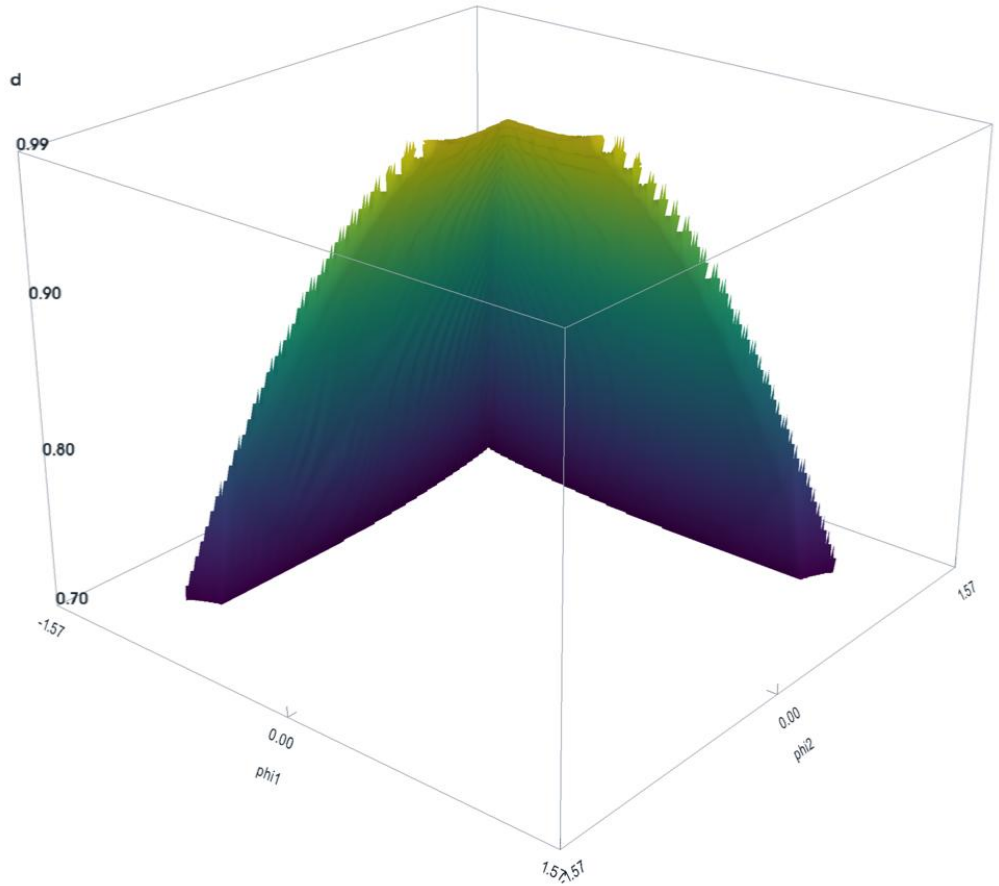


Red: snap edges, where p changes sign with a jump. Black: smooth crossings through the S-shape.

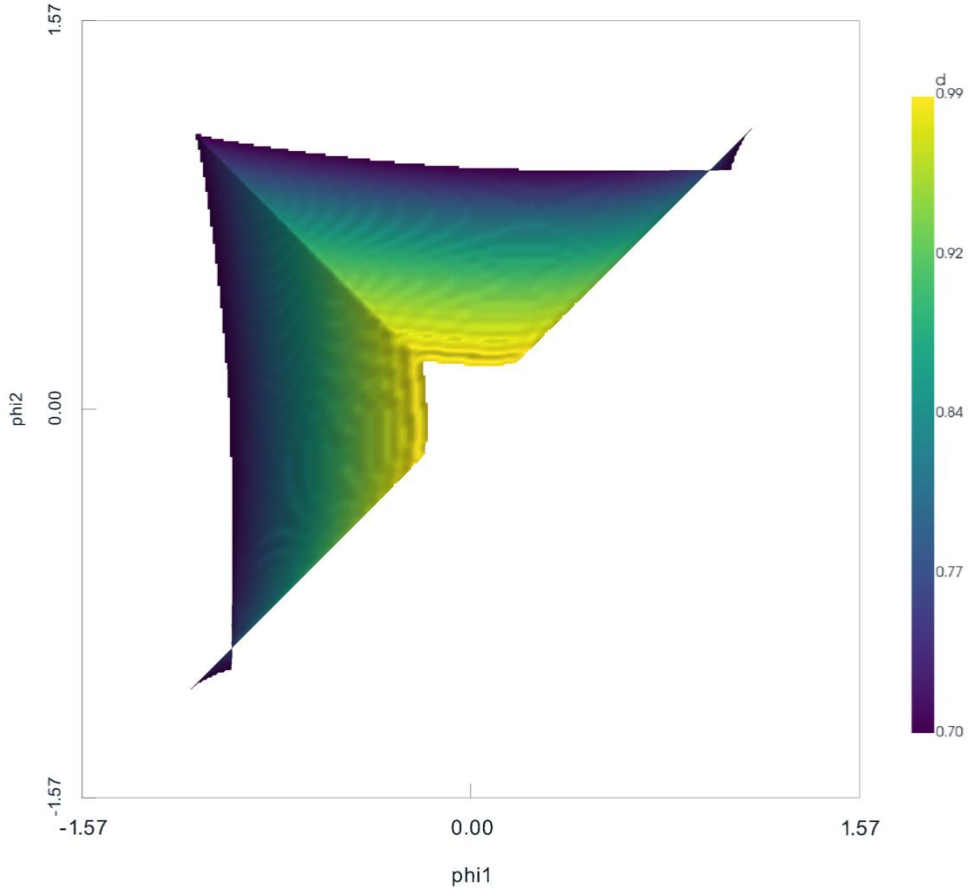


As d → 1 the L closes in: at $\varphi_2 = 0$ the snap sits at $\varphi_1 \approx -0.98$ (d = 0.70), −0.57 (0.90) and −0.18 (0.99).

The snap surface in 3D

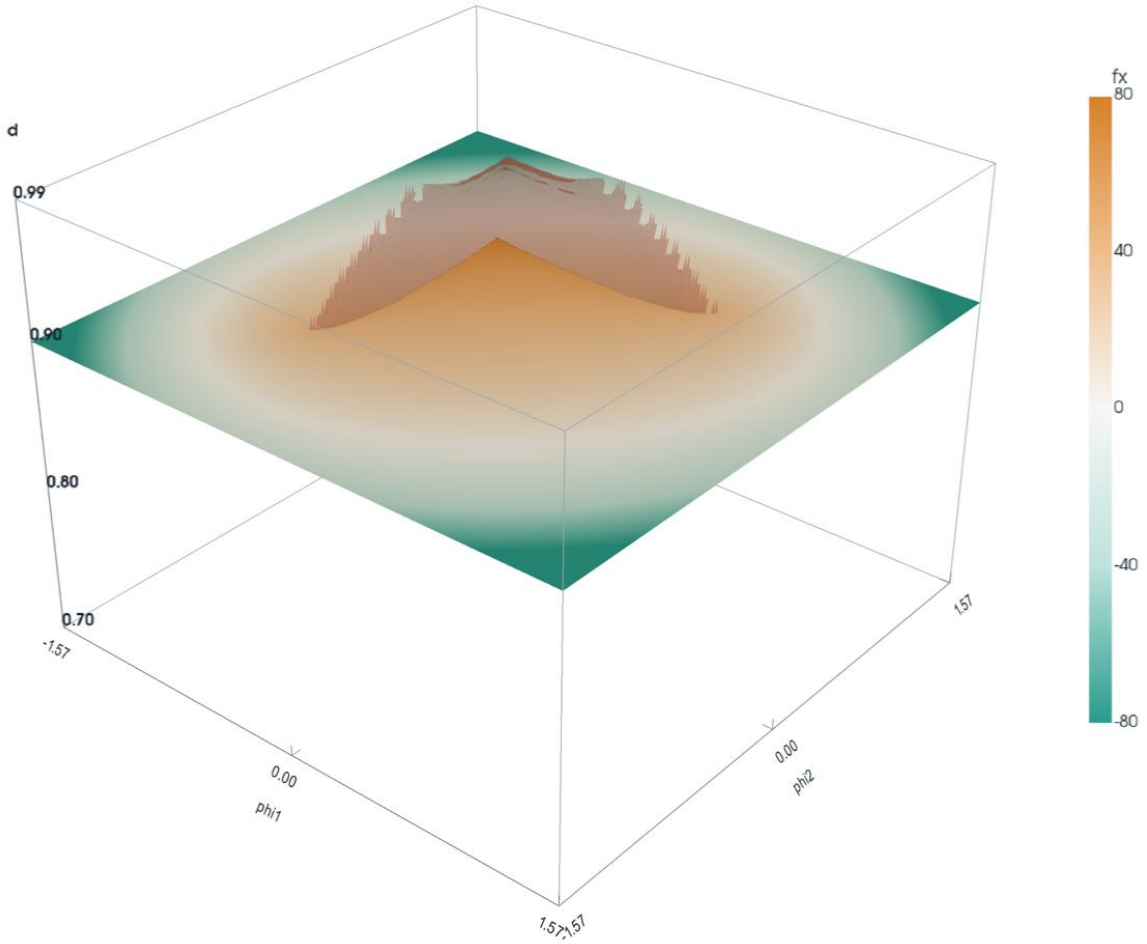


Oblique view, d stretched ×8. The p = 0 contour of the 17.2 M-tetrahedron grid, kept where p jumps: 29,756 snap edges, area 2.62.



Down the d axis: every layer is the L-shape, and the sheet closes in on the origin as d → 1. Colour: d.

The sheet rises from the step in axial force



Layer d = 0.90 coloured by f_x (clipped at ±80); the snap sheet above it is translucent.

Across the snap at $\phi_2 = 0$, f_x roughly doubles at every d:

30.1 → 61.5 d = 0.80, jump at $\phi_1 \approx -0.80$

28.3 → 58.7 d = 0.90, jump at $\phi_1 \approx -0.57$

27.4 → 55.4 d = 0.97, jump at $\phi_1 \approx -0.31$

The surface is built from the shape alone, through the sign of p, yet it lands exactly on the step the forces see.

Continuation strategy

How the solver walks the parameter space and guards against branch switches

Traverse the grid; reject jumps and unstable results

Cold-start once at $\phi_1 = \phi_2 = 0$, smallest d; visit nodes by grid distance from it

→

Warm-start θ, f_x, f_y from a solved grid neighbour, nearest first

→

Newton solve (SNES, at most 50 iterations)

→

No jump in q, and stable?

yes →

Accept, store, and write the mirror point

no ↓

Rejected or diverged: try the next neighbour. After 3 failures from one branch, wait for the other; a waiting node retries whenever a neighbour is solved.

Grid

Rectilinear (d, ϕ_1, ϕ_2) grid split into 6 tetrahedra per cell; warm starts come from the 26 grid neighbours

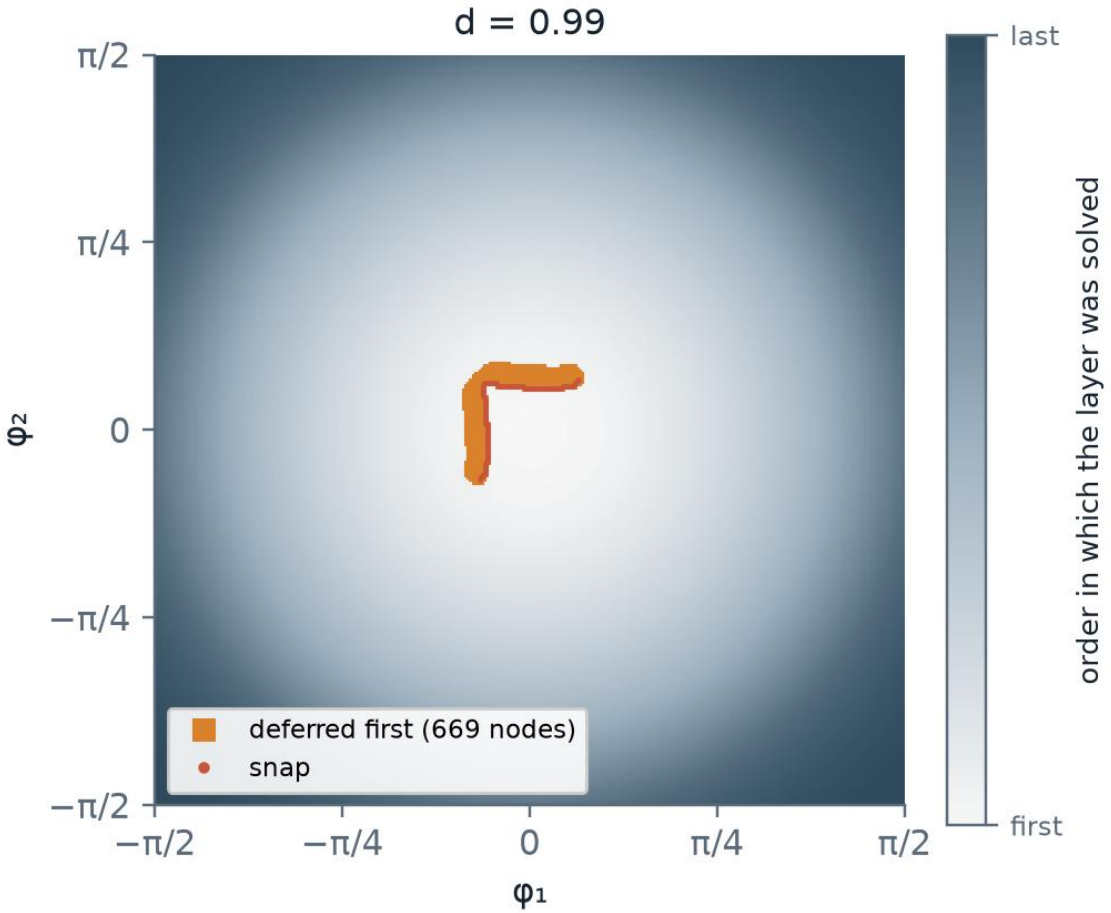
Restartable

Live NPZ checkpoints, also on Ctrl-C; a resume reloads every stored θ as a warm-start source

Half domain

Only $\phi_1 + \phi_2 \geq 0$ is solved; a precomputed mirror index fills the other half

Going around a fold: defer, never jump



d = 0.99 layer: the front grows outward; orange nodes had to wait for a neighbour on the other branch.

1. **Defer**
   A node with no acceptable warm start is set aside, not forced; it is retried each time a new neighbour is solved.

2. **Wait for the other branch**
   After 3 failures from neighbours on one branch it only tries the other, so a node past a fold waits for the far side.

3. **The long way round**
   The front reaches the cut-off region through the smooth crossings and approaches it from the far side.

4. **Last resort**
   Only if the whole front stalls is the nearest waiting node cold-started.

Production run: 1,155 nodes waited (all at d ≥ 0.97); one extra cold start, at (0.99, 0.20, 0.20).

A stability test catches what the jump test misses

Constrained second variation

$$Q[v] = \int_0^L v'^{\,2} - (f_x \cos\theta + f_y \sin\theta)\, v^2 \, ds$$

$$v(0) = v(L) = 0, \quad \int_0^L v \cos\theta \, ds = \int_0^L v \sin\theta \, ds = 0$$

$$n_- = \mathrm{neg}(A) + \mathrm{pos}(C^T A^{-1} C) - 2$$

Stable when $n_- = 0$: no direction lowers the energy.

17

unstable results rejected

during the production run

0

unstable in the final map

all 2,976,750 nodes re-tested

1

Why the jump test is not enough

On $\phi_1 = -\phi_2$ the rod is symmetric ($f_y = 0$); the bent-up branch loses stability at a symmetry-breaking bifurcation, not a fold. Newton follows it smoothly, with no jump in q.

2

Inertia of a bordered matrix

A is the tridiagonal P1 form, C the two constraint vectors (Haynsworth). One LDL^T sweep: O(N), about 0.2 ms against an 11 ms solve.

3

Every candidate

Warm starts and cold starts alike: an unstable result counts as a failure and the node moves on to the next neighbour.

Deriving the exact first mode used for the cold start

1. $\phi_1 = \phi_2 = 0$ is mirror-symmetric, so $f_y = 0$ and the elastica is a pendulum

$$\theta'' + \lambda^2 \sin\theta = 0, \quad \lambda^2 = f_x, \quad f_y = 0$$

2. Multiply by θ′ and integrate once; θ′ = 0 at the peak angle α

$$\theta'^2 = 4\lambda^2\left(k^2 - \sin^2\frac{\theta}{2}\right), \qquad k = \sin\frac{\alpha}{2}$$

3. Substituting $\sin(\theta/2) = k \sin\psi$ gives $\psi' = \lambda\sqrt{(1 - m\sin^2\psi)}$, so $\sin\psi = \mathrm{sn}(\lambda s \mid m)$

$$\theta(s) = 2\sin^{-1}[k\,\mathrm{sn}(\lambda s \mid m)], \qquad m = k^2$$

4. θ(L) = 0 and $\int \sin\theta\, ds = 0$ need one full period of sn (4K): four quarter-waves

$$\lambda L = 4K(m) \quad \Rightarrow \quad f_x = \left(\frac{4K(m)}{L}\right)^2$$

5. The end distance fixes m; d/L falls monotonically in m, so a 1-D root-find is safe

$$\frac{d}{L} = \frac{1}{L}\int_0^L \cos\theta\; ds = \frac{2E(m)}{K(m)} - 1$$

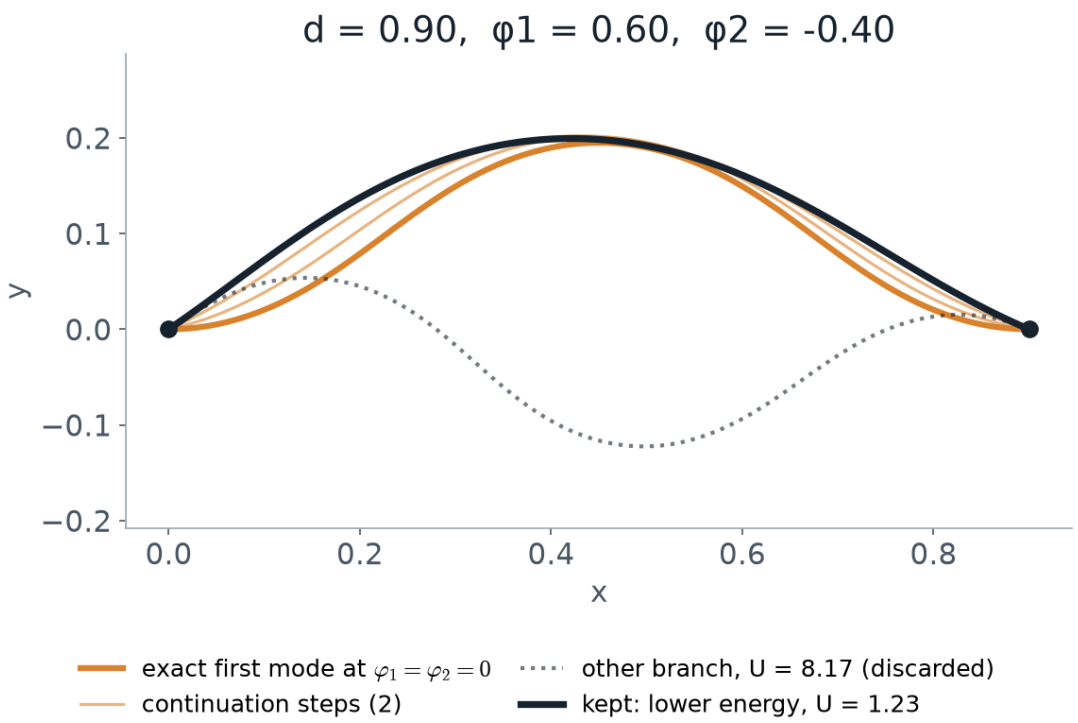
6. Small bow, $m \to 0$: the $A \sin(2\pi s/L)$ shape with $A = 2\sqrt{(1 - d/L)}$, and the Euler load

$$\theta \approx 2\sqrt{m}\,\sin\frac{2\pi s}{L}, \quad m \approx 1 - \frac{d}{L}, \quad f_x \to \frac{4\pi^2}{L^2}$$

$$K(m) = \int_0^{\pi/2} \frac{d\psi}{\sqrt{1 - m\sin^2\psi}}, \qquad E(m) = \int_0^{\pi/2} \sqrt{1 - m\sin^2\psi}\; d\psi$$

Exact first-mode cold start, then continue the clamp angles

$$\theta \approx 2\sqrt{m}\,\sin\frac{2\pi s}{L}, \quad m \approx 1-\frac{d}{L}, \quad f_x \to \frac{4\pi^2}{L^2}$$



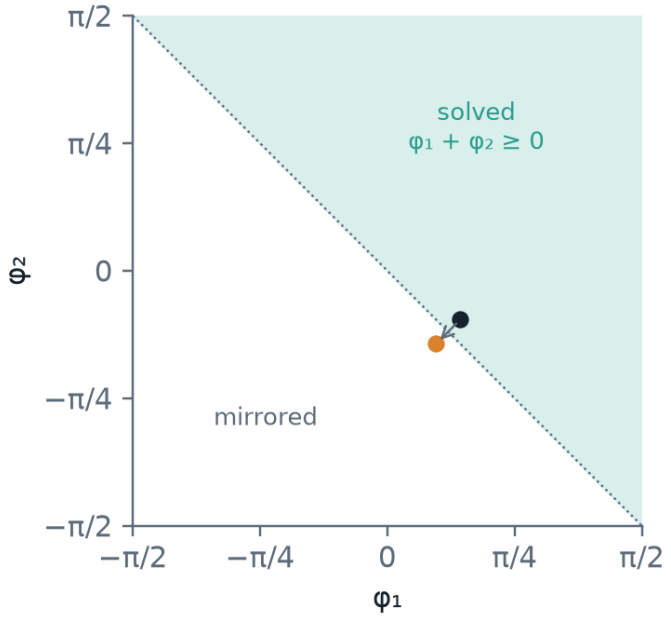
Predictor for each continuation step

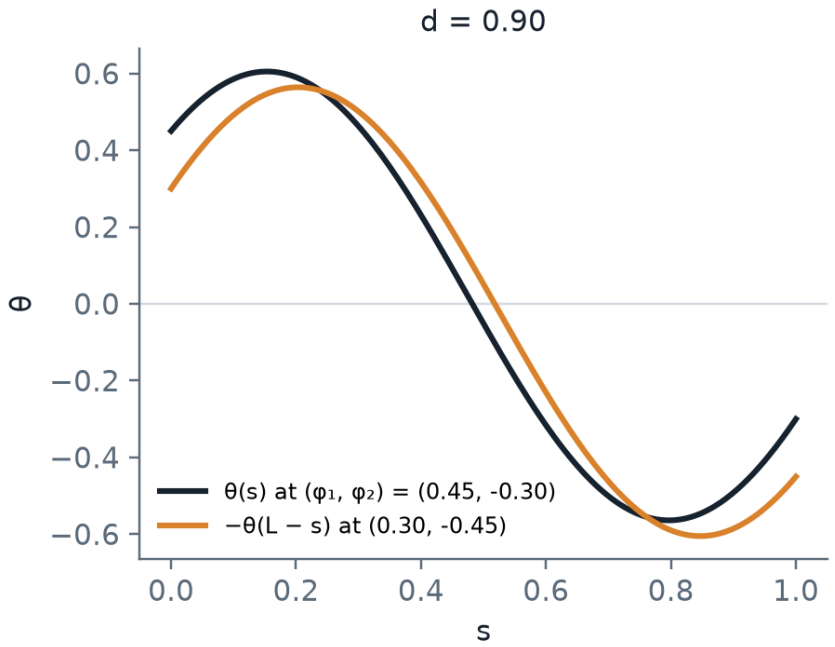
$$\theta_0^{(k+1)}(s) = \theta^{(k)}(s) + \Delta\varphi_1\left(1-\frac{s}{L}\right) + \Delta\varphi_2\frac{s}{L}$$

1. Exact at $\phi_1 = \phi_2 = 0$ for any d, with $\int \sin\theta\, ds = 0$: Newton converges in 2 iterations
2. Steps ≤ 0.25 rad, ≤ 15 Newton iterations each; a failed step halves, giving up below 0.02 rad (a fold)
3. Both hump signs are continued and the lower energy kept, so the result does not depend on which branch the ramp follows

Mirror symmetry halves the number of solves

$$(d, \varphi_1, \varphi_2) \mapsto (d, -\varphi_2, -\varphi_1): \quad \tilde{\theta}(s) = -\theta(L - s), \quad \tilde{f}_x = f_x, \quad \tilde{f}_y = -f_y, \quad \tilde{M}_1 = M_2, \quad \tilde{p} = p$$





1. Reflecting the rod about mid-span maps (ϕ_1, ϕ_2) to $(-\phi_2, -\phi_1)$

2. Only $\phi_1 + \phi_2 \geq 0$ is solved; points on the anti-diagonal are solved once

3. f_x, U and p carry over, f_y flips sign, and the left and right moments swap

Final run

$d \in [0.70, 0.99]$, $\phi_1, \phi_2 \in [-\pi/2, \pi/2]$, $\Delta = 0.01$: 2.98 M equilibria and their level sets

2.98 M equilibria, every one checked

2,976,750

equilibria

30 × 315 × 315 grid; 1.49 M Newton solves, the mirror fills the rest

≈ 4 h

on one core

about 11 ms per solve, from one cold start (plus one more near d = 1)

0.27 %

max error vs. BVP

300 random nodes re-solved with scipy solve_bvp; median 0.014 %

0

unstable equilibria

stability test on all 2,976,750 nodes

Exact checks pass

Mirror pairs agree to 3×10^{-8}. At $\phi_1 = \phi_2 = 0$ every layer matches the exact first mode: f_x within 1.3×10^{-4}.

Branch share

Bent up on 70.0 % of the layer at d = 0.70, falling almost linearly to 50.5 % at d = 0.99. Snap surface area 2.62 (29,756 snap edges).

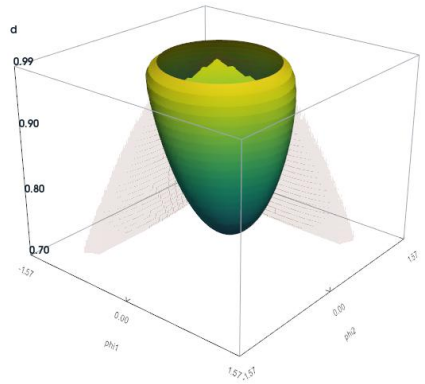
Ranges

f_x from −13,785 (tension, d = 0.99, $|\phi| = \pi/2$) to +92; $|f_y| \le 340$; $|M| \le 168$; U from 0.10 to 142.

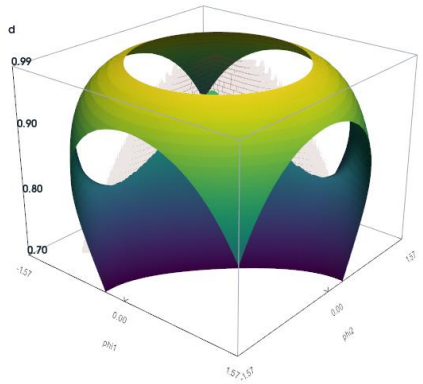
ParaView: data/production/post/elastica_solution_space.vtu (every node, all fields).

Constant-energy surfaces: shells around the straight rod

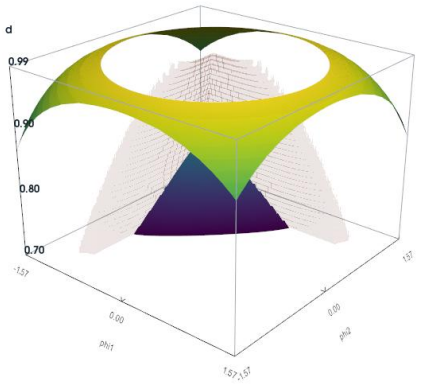
U = 2



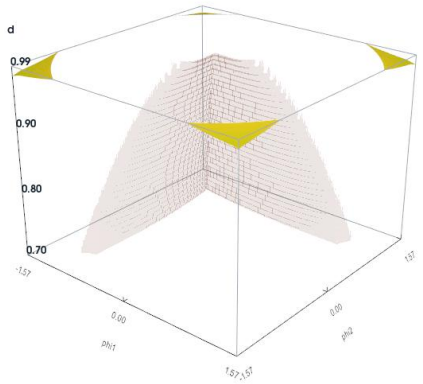
U = 8



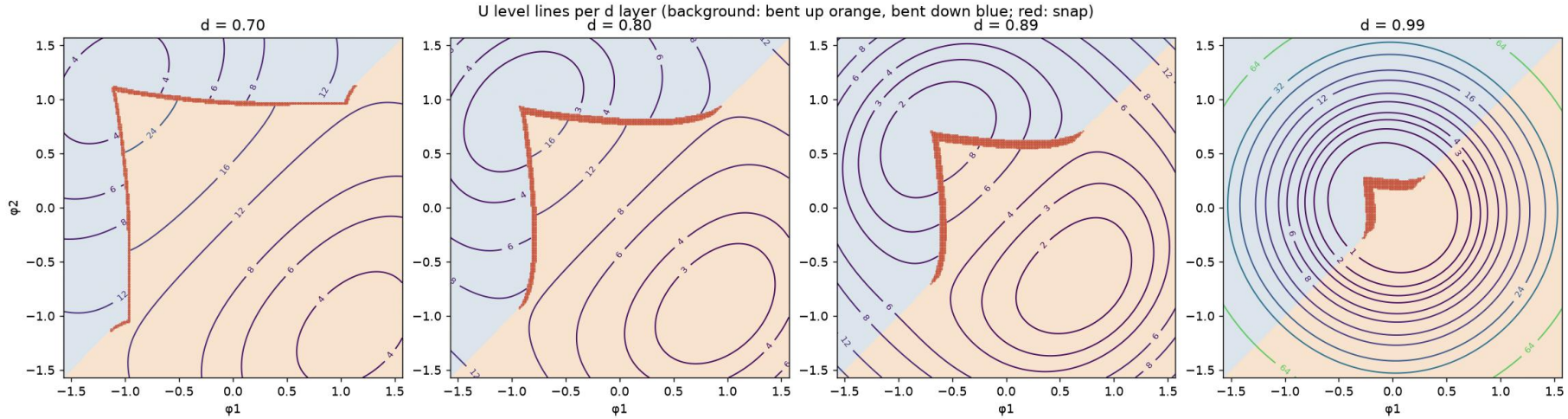
U = 16



U = 64



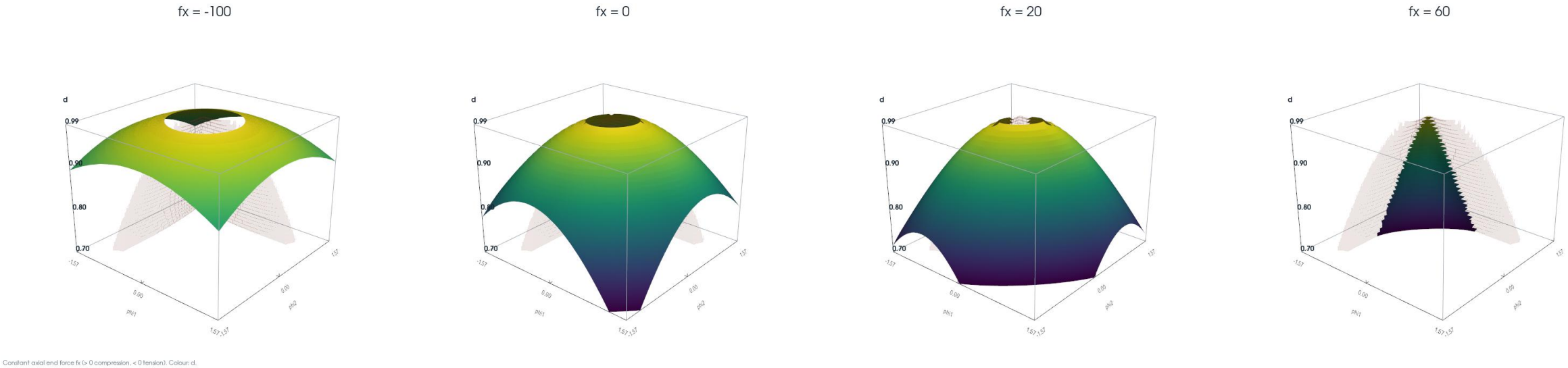
Constant bending energy U = 1/2 ∫ θ'^2 ds (θ = L = 1). Colour: d. Faint red: snap part of the ρ = 0 surface; level sets stop at it.

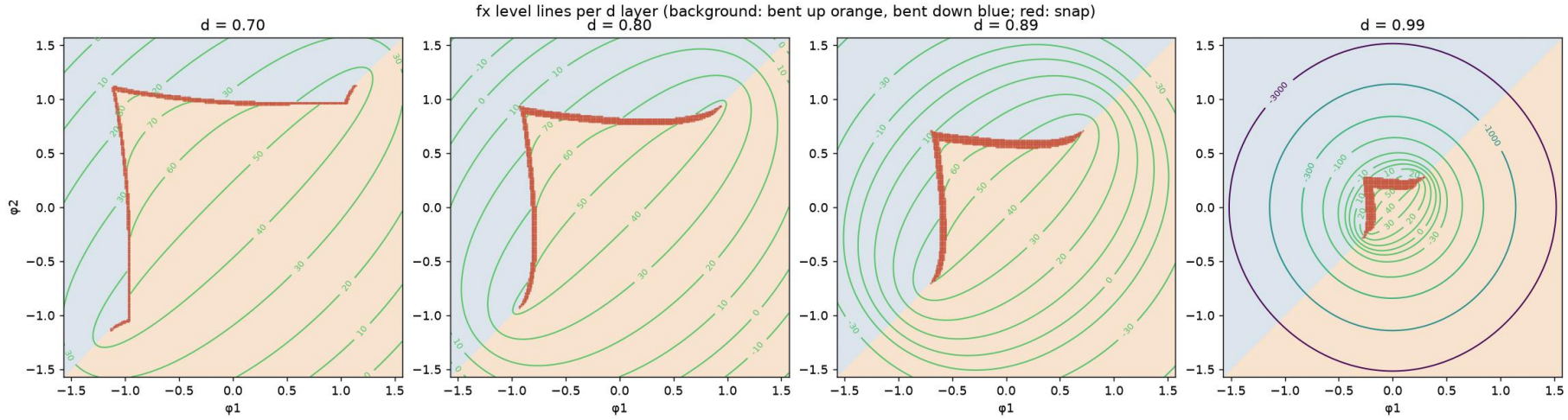


$U = \frac{1}{2}\int \theta'^2\, ds$

From 0.10 to 142. Lowest for a nearly straight rod ($d \to 1$, $\phi \to 0$): U = 2 is a bowl around that corner and higher levels are wider domes. Each ends at the snap (faint red), where U jumps.

Constant axial force f_x: bounded in compression

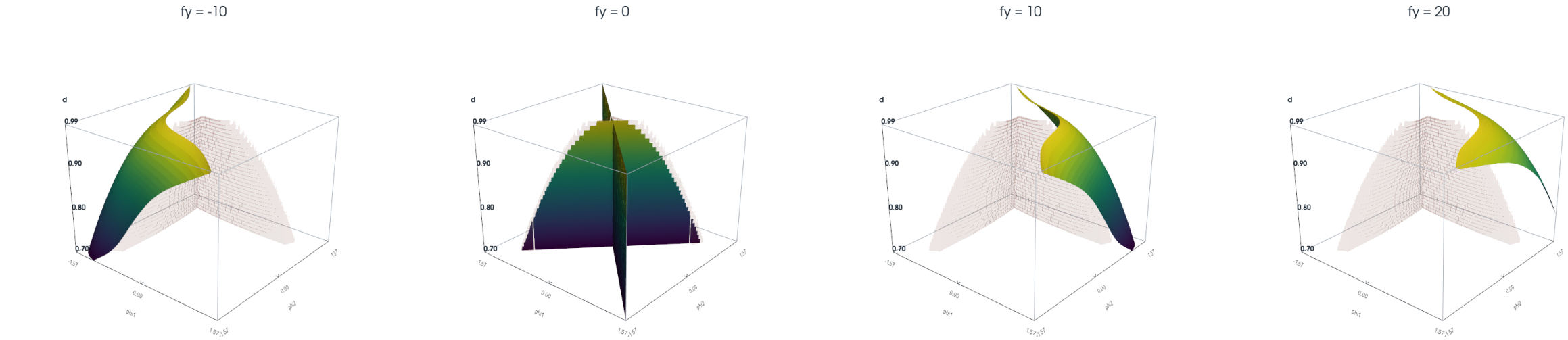




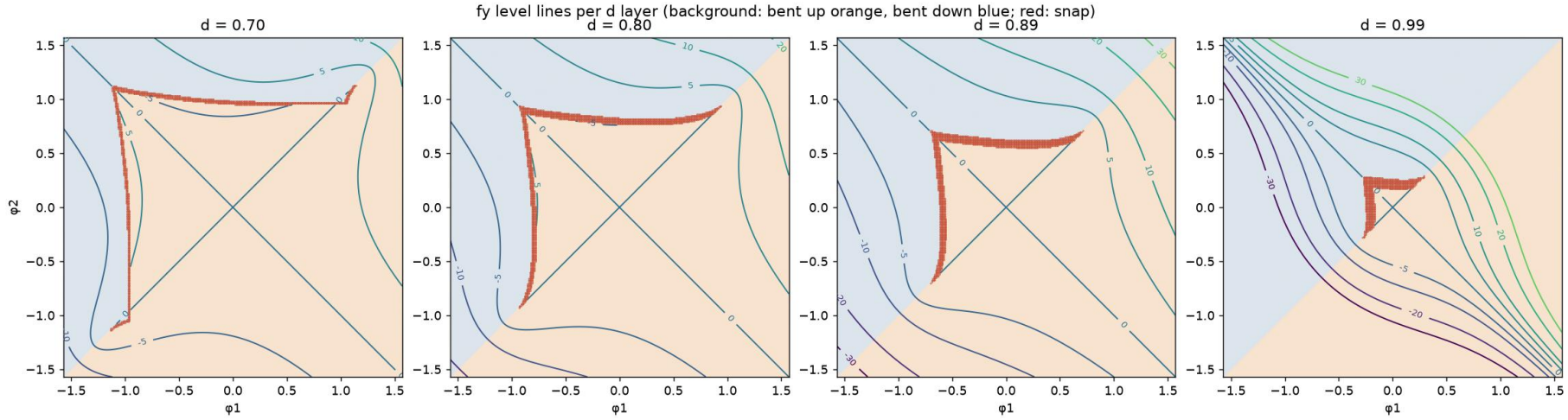
$f_x > 0$: compression

From −13,785 (tension, d = 0.99, $|\phi| = \pi/2$) to +92. Compressive levels exist only inside the bent-up region: f_x = 60 is a sliver and nothing reaches 100.

Constant transverse force f_y: mirror-image pairs



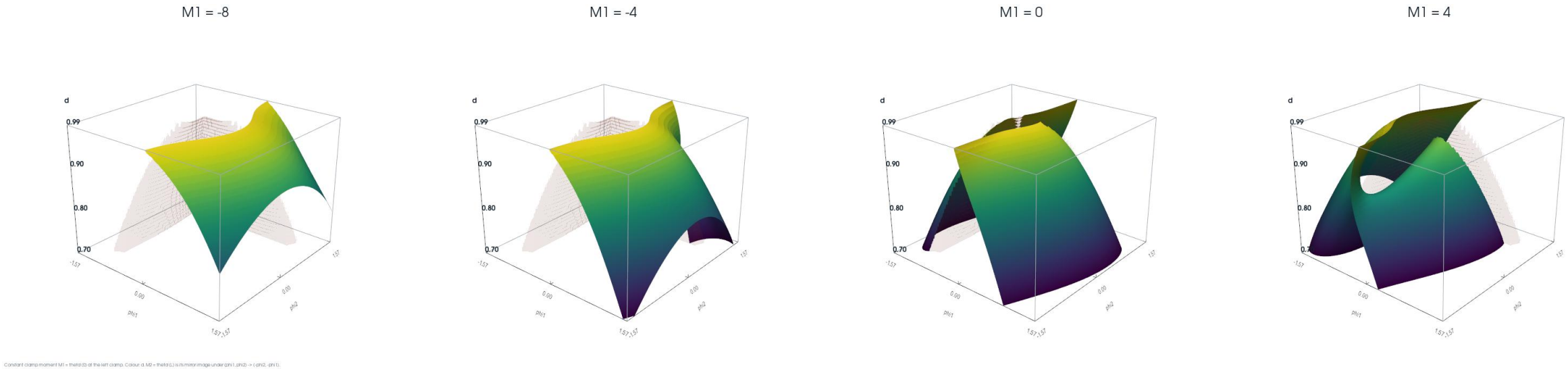
Constant transverse end force fy. Colour: d.

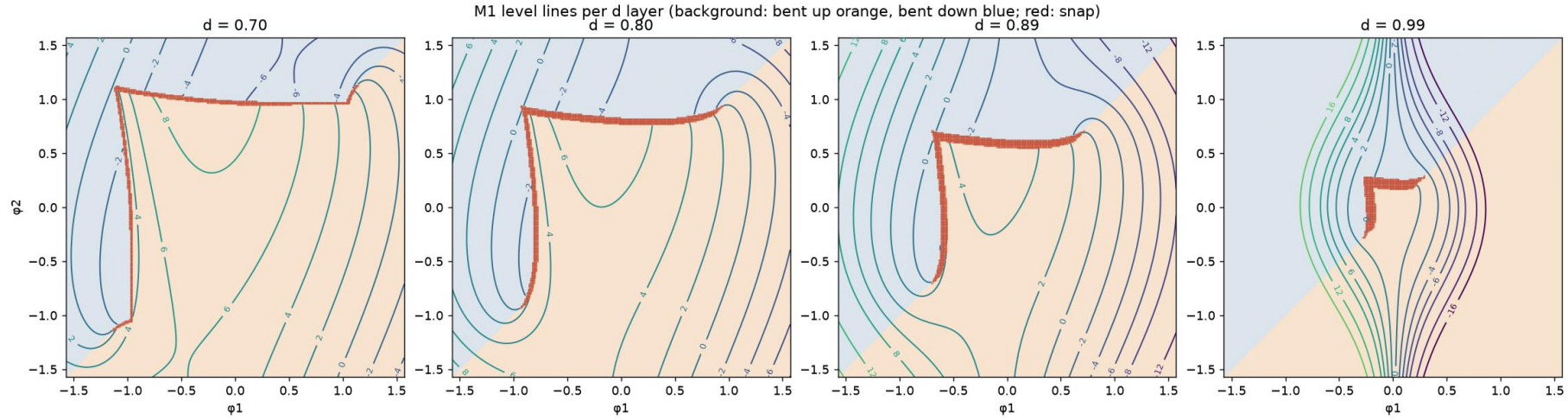


Odd under the mirror

$|f_y| \le 340$. $(\phi_1, \phi_2) \to (-\phi_2, -\phi_1)$ flips f_y, so $f_y = 0$ holds on the whole plane $\phi_1 = -\phi_2$ and ± levels come in pairs of equal area.

Constant clamp moment M_1; M_2 is its mirror image





$M_1 = \theta'(0)$

Within ±168. M_2 at (ϕ_1, ϕ_2) is M_1 at $(-\phi_2, -\phi_1)$, so its surfaces are mirror images (both in the VTU). Small $|M_1|$ levels split into two sheets.